Cover Letter

Manuscript Title: WaveletMask: Wavelet-Domain Mask-Guided Degradation Detection for Old Film Restoration

Journal: Applied Sciences (MDPI)

Submission type: Resubmission after minor revision (Round 2)

Previous Manuscript ID: applsci-4348902

Dear Editor,

We thank you and the three reviewers for the constructive re-evaluation of our manuscript. This submission addresses the remaining minor comments from the second round of review for manuscript applsci-4348902. No new experiments were required; the changes are limited to additional transparency disclosures, minor wording qualifications, and a small number of editorial clarifications. A point-by-point response to each reviewer is provided in the accompanying response documents, and all changes are marked in the highlighted version of the manuscript. Below we summarize the principal revisions.

Principal Revisions

1. **Unified-benchmark transparency (Reviewer 1, N1).** We added a sentence in Appendix D stating that all external methods in the unified benchmark were trained with their officially recommended configurations under a common iteration budget, and that the corresponding per-method configuration files will be released with the code and training materials described in the Data Availability statement.

2. **Subjective-study visibility (Reviewer 1, N2).** We added a specific anchor sentence in Section 4.4.2 pointing readers to the expert-level subjective study with 34 trained raters reported in the Supplementary Materials (Tables S1–S2 and Figure S4).

3. **Conservative no-reference claim (Reviewer 1, N3).** We qualified the conclusion to state that WaveletMask is the only method placing within the top two on all three no-reference metrics, narrowly on CLIPIQA+ where it leads the next-best method by 0.0005.

4. **Baseline snapshot disclosure (Reviewer 1, N4).** We added an explicit sentence in Section 4.4.1 clarifying that the RTN paired snapshot and the unified-benchmark snapshot come from different training checkpoints and metric configurations, so the unified-benchmark RTN row should not be read as the paired baseline.

5. **LHFM proxy limitation (Reviewer 2, E1).** We added one sentence in Appendix C.3 explaining that the high-pass residual proxy cannot

distinguish true scratches from texture edges, which partly explains the local branch's below-baseline top-5% hit rate on this proxy while it still improves end-to-end restoration.

6. **Runtime protocol wording (Reviewer 2, E2).** We replaced the colloquial runtime phrasing in Appendix D with an explicit disclosure that batch-4 inference at 1280×720 exceeded the available 48 GB L40 GPU memory.

7. **Fourier-temporal future direction (Reviewer 3, N1).** We added an explicit future-direction sentence in Section 5.3 naming Fourier-temporal end-to-end retraining as a concrete next step for closing the frozen-Haar gap.

All changes are confined to wording, disclosures, and editorial clarifications; no tables, figures, or statistical results were altered. The highlighted version of the manuscript marks every change relative to the previous round.

We believe these minor revisions fully address the reviewers' remaining concerns and thank you again for the opportunity to revise.

Ethics Statement

The expert subjective evaluation referenced in the manuscript was an internal visual-quality scoring of restored images by internal expert raters. All algorithmic experiments were conducted on publicly available video datasets for old-film restoration evaluation. The internal scoring did not recruit external human subjects, collect human biological samples, record identifiable personal or private information, or involve any intervention on the human body. In accordance with Article 2 of the *Measures for Scientific and Technological Ethics Review (Trial)* (Guo Ke Fa Jian [2023] No. 167; jointly issued by the Ministry of Science and Technology and nine other Chinese authorities), this work does not fall within the scope requiring mandatory ethics review, and no ethics approval or exemption document is required.

Declarations

We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with, or published in, another journal. All authors have approved the manuscript and agree with its submission to Applied Sciences. We acknowledge that this work was previously submitted to Applied Sciences (Manuscript ID: applsci-4348902), and the present resubmission addresses all reviewer and editor comments from that round.

Sincerely,

The Authors